

Homework 6

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Instructions

Answer the following questions and/or complete the exercises in RMarkdown. Please embed all of your code and push the final work to your repository. Your report should be organized, clean, and run free from errors. Remember, you must remove the `#` for any included code chunks to run.

Load the tidyverse

```
library("tidyverse")
library("janitor")
```

Data

For this assignment, we will use data from a study on vertebrate community composition and impacts from defaunation in Gabon, Africa. One thing to notice is that the data include 24 separate transects. Each transect represents a path through different forest management areas.

Reference: Koerner SE, Poulsen JR, Blanchard EJ, Okouyi J, Clark CJ. Vertebrate community composition and diversity declines along a defaunation gradient radiating from rural villages in Gabon. *Journal of Applied Ecology*. 2016. A description of the variables is included inside the data folder.

1. Load `IvindoData_DryadVersion.csv` and store it as a new object called `gabon`.

```
gabon<- read_csv("data/IvindoData_DryadVersion.csv")%>% clean_names()

## Rows: 24 Columns: 26
## -- Column specification -----
## Delimiter: ","
## chr (2): HuntCat, LandUse
## dbl (24): TransectID, Distance, NumHouseholds, Veg_Rich, Veg_Stems, Veg_lian...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

2. Use one or more of the summary functions you have learned to get an idea of the structure of the data.

```
glimpse(gabon)
```

```
## Rows: 24
## Columns: 26
## $ transect_id      <dbl> 1, 2, 2, 3, 4, 5, 6, 7, 8, 9, 13, 14, 15, 16, ~
## $ distance         <dbl> 7.14, 17.31, 18.32, 20.85, 15.95, 17.47, 24.0~
## $ hunt_cat         <chr> "Moderate", "None", "None", "None", "None", "~
## $ num_households   <dbl> 54, 54, 29, 29, 29, 29, 29, 54, 25, 73, 46, 5~
## $ land_use         <chr> "Park", "Park", "Park", "Logging", "Park", "P~
## $ veg_rich         <dbl> 16.67, 15.75, 16.88, 12.44, 17.13, 16.50, 14.~
## $ veg_stems        <dbl> 31.20, 37.44, 32.33, 29.39, 36.00, 29.22, 31.~
## $ veg_liana        <dbl> 5.78, 13.25, 4.75, 9.78, 13.25, 12.88, 8.38, ~
## $ veg_dbh          <dbl> 49.57, 34.59, 42.82, 36.62, 41.52, 44.07, 51.~
## $ veg_canopy       <dbl> 3.78, 3.75, 3.43, 3.75, 3.88, 2.50, 4.00, 4.0~
## $ veg_understory   <dbl> 2.89, 3.88, 3.00, 2.75, 3.25, 3.00, 2.38, 2.7~
## $ ra_apes          <dbl> 1.87, 0.00, 4.49, 12.93, 0.00, 2.48, 3.78, 6.~
## $ ra_birds         <dbl> 52.66, 52.17, 37.44, 59.29, 52.62, 38.64, 42.~
## $ ra_elephant      <dbl> 0.00, 0.86, 1.33, 0.56, 1.00, 0.00, 1.11, 0.4~
## $ ra_monkeys       <dbl> 38.59, 28.53, 41.82, 19.85, 41.34, 43.29, 46.~
## $ ra_rodent        <dbl> 4.22, 6.04, 1.06, 3.66, 2.52, 1.83, 3.10, 1.2~
## $ ra_ungulate      <dbl> 2.66, 12.41, 13.86, 3.71, 2.53, 13.75, 3.10, ~
## $ rich_all_species <dbl> 22, 20, 22, 19, 20, 22, 23, 19, 19, 19, 21, 2~
## $ evenness_all_species <dbl> 0.793, 0.773, 0.740, 0.681, 0.811, 0.786, 0.8~
## $ diversity_all_species <dbl> 2.452, 2.314, 2.288, 2.006, 2.431, 2.429, 2.5~
## $ rich_bird_species <dbl> 11, 10, 11, 8, 8, 10, 11, 11, 11, 9, 11, 11, ~
## $ evenness_bird_species <dbl> 0.732, 0.704, 0.688, 0.559, 0.799, 0.771, 0.8~
## $ diversity_bird_species <dbl> 1.756, 1.620, 1.649, 1.162, 1.660, 1.775, 1.9~
## $ rich_mammal_species <dbl> 11, 10, 11, 11, 12, 12, 12, 8, 8, 10, 10, 11, ~
## $ evenness_mammal_species <dbl> 0.736, 0.705, 0.650, 0.619, 0.736, 0.694, 0.7~
## $ diversity_mammal_species <dbl> 1.764, 1.624, 1.558, 1.484, 1.829, 1.725, 1.9~
```

3. Clean the data as necessary following the conventions from lab 6.

```
gabon<- gabon%>%
  clean_names()
glimpse(gabon)
```

```
## Rows: 24
## Columns: 26
## $ transect_id      <dbl> 1, 2, 2, 3, 4, 5, 6, 7, 8, 9, 13, 14, 15, 16, ~
## $ distance         <dbl> 7.14, 17.31, 18.32, 20.85, 15.95, 17.47, 24.0~
## $ hunt_cat         <chr> "Moderate", "None", "None", "None", "None", "~
## $ num_households   <dbl> 54, 54, 29, 29, 29, 29, 29, 54, 25, 73, 46, 5~
## $ land_use         <chr> "Park", "Park", "Park", "Logging", "Park", "P~
## $ veg_rich         <dbl> 16.67, 15.75, 16.88, 12.44, 17.13, 16.50, 14.~
## $ veg_stems        <dbl> 31.20, 37.44, 32.33, 29.39, 36.00, 29.22, 31.~
## $ veg_liana        <dbl> 5.78, 13.25, 4.75, 9.78, 13.25, 12.88, 8.38, ~
## $ veg_dbh          <dbl> 49.57, 34.59, 42.82, 36.62, 41.52, 44.07, 51.~
## $ veg_canopy       <dbl> 3.78, 3.75, 3.43, 3.75, 3.88, 2.50, 4.00, 4.0~
## $ veg_understory   <dbl> 2.89, 3.88, 3.00, 2.75, 3.25, 3.00, 2.38, 2.7~
## $ ra_apes          <dbl> 1.87, 0.00, 4.49, 12.93, 0.00, 2.48, 3.78, 6.~
## $ ra_birds         <dbl> 52.66, 52.17, 37.44, 59.29, 52.62, 38.64, 42.~
```

```
## $ ra_elephant      <dbl> 0.00, 0.86, 1.33, 0.56, 1.00, 0.00, 1.11, 0.4~
## $ ra_monkeys       <dbl> 38.59, 28.53, 41.82, 19.85, 41.34, 43.29, 46.~
## $ ra_rodent        <dbl> 4.22, 6.04, 1.06, 3.66, 2.52, 1.83, 3.10, 1.2~
## $ ra_ungulate      <dbl> 2.66, 12.41, 13.86, 3.71, 2.53, 13.75, 3.10, ~
## $ rich_all_species <dbl> 22, 20, 22, 19, 20, 22, 23, 19, 19, 19, 21, 2~
## $ evenness_all_species <dbl> 0.793, 0.773, 0.740, 0.681, 0.811, 0.786, 0.8~
## $ diversity_all_species <dbl> 2.452, 2.314, 2.288, 2.006, 2.431, 2.429, 2.5~
## $ rich_bird_species <dbl> 11, 10, 11, 8, 8, 10, 11, 11, 11, 9, 11, 11, ~
## $ evenness_bird_species <dbl> 0.732, 0.704, 0.688, 0.559, 0.799, 0.771, 0.8~
## $ diversity_bird_species <dbl> 1.756, 1.620, 1.649, 1.162, 1.660, 1.775, 1.9~
## $ rich_mammal_species <dbl> 11, 10, 11, 11, 12, 12, 12, 8, 8, 10, 10, 11,~
## $ evenness_mammal_species <dbl> 0.736, 0.705, 0.650, 0.619, 0.736, 0.694, 0.7~
## $ diversity_mammal_species <dbl> 1.764, 1.624, 1.558, 1.484, 1.829, 1.725, 1.9~
```

4. Use `mutate()` Change the variables `HuntCat`, `LandUse`, and `TransectID` to factors. Be sure to store the output as a new dataframe and use it for the remaining questions.

```
?factor
```

```
## starting httpd help server ... done
```

```
gabon%>%
  mutate(hunt_cat=as.factor(hunt_cat),
         land_use=as.factor(land_use),
         transect_id=as.factor(transect_id))
```

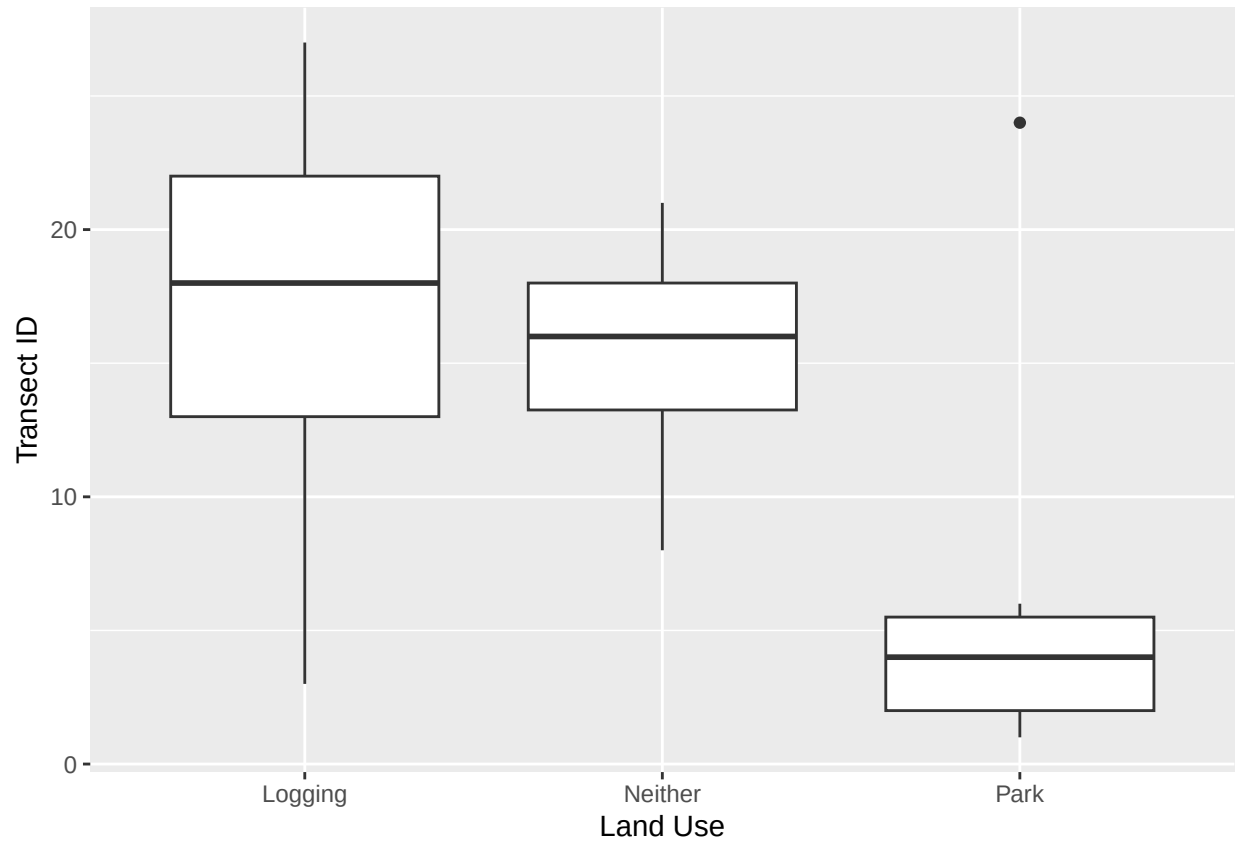
```
## # A tibble: 24 x 26
##   transect_id distance hunt_cat num_households land_use veg_rich veg_stems
##   <fct>          <dbl> <fct>          <dbl> <fct>          <dbl>    <dbl>
## 1 1              7.14 Moderate          54 Park          16.7      31.2
## 2 2             17.3 None              54 Park          15.8      37.4
## 3 2             18.3 None              29 Park          16.9      32.3
## 4 3             20.8 None              29 Logging        12.4      29.4
## 5 4             16.0 None              29 Park          17.1       36
## 6 5             17.5 None              29 Park          16.5      29.2
## 7 6             24.1 None              29 Park          14.8      31.2
## 8 7             19.8 None              54 Logging        13.2      32.6
## 9 8              5.78 High              25 Neither        12.6      23.7
## 10 9             5.13 High              73 Logging         16      27.1
## # i 14 more rows
## # i 19 more variables: veg_liana <dbl>, veg_dbh <dbl>, veg_canopy <dbl>,
## #   veg_understory <dbl>, ra_apes <dbl>, ra_birds <dbl>, ra_elephant <dbl>,
## #   ra_monkeys <dbl>, ra_rodent <dbl>, ra_ungulate <dbl>,
## #   rich_all_species <dbl>, evenness_all_species <dbl>,
## #   diversity_all_species <dbl>, rich_bird_species <dbl>,
## #   evenness_bird_species <dbl>, diversity_bird_species <dbl>, ...
```

5. Make a plot that shows the number of transects for each land use type.

```
#gabon%>%
#  #distinct(transect_id, land_use)%>%
#  #ggplot(aes(x=land_use))+
```

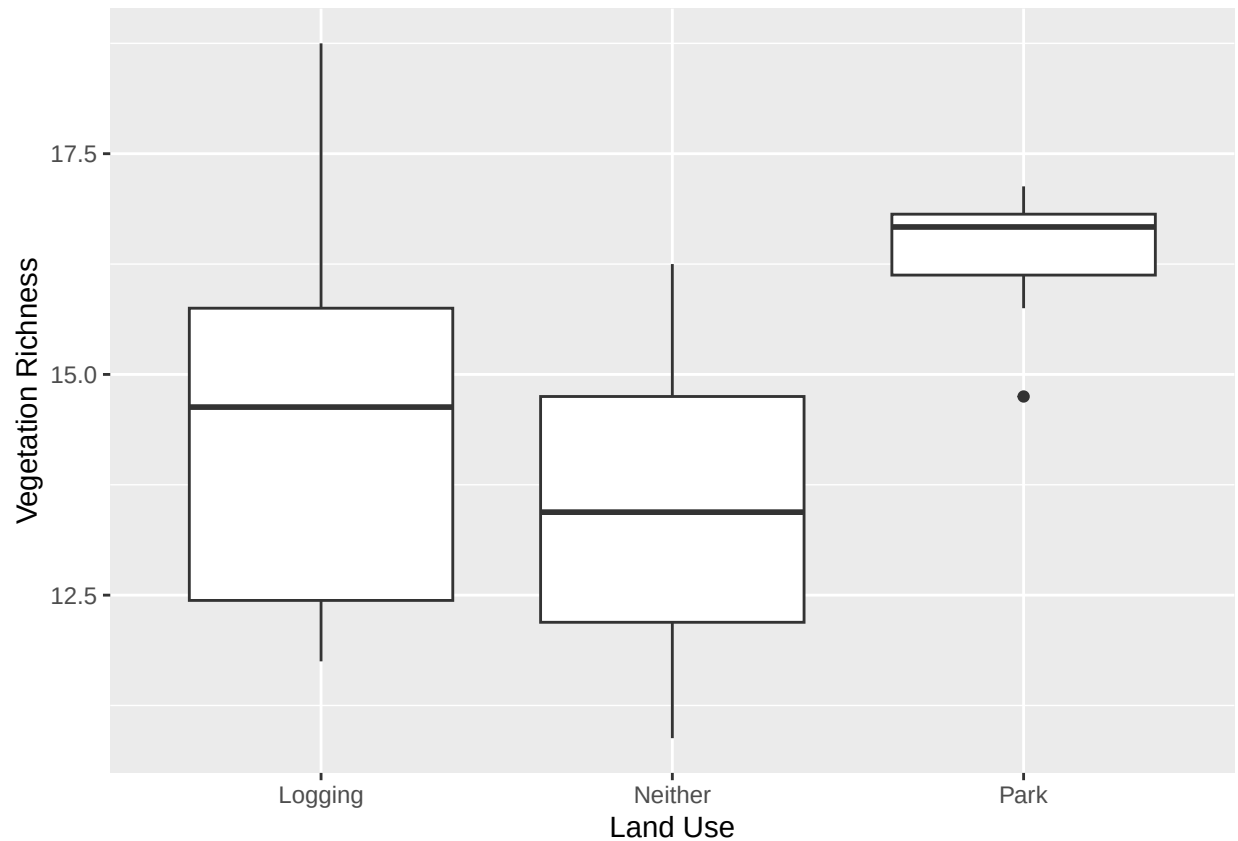
```
#geom_bar()

ggplot(data=gabon,
       mapping=aes(x=land_use,y= transect_id))+
  geom_boxplot()+
  labs(x= "Land Use", y= "Transect ID")
```



6. How does vegetation richness compare among land use types?

```
ggplot(data=gabon,
       mapping=aes(x=land_use,y= veg_rich))+
  geom_boxplot()+
  labs(x= "Land Use", y= "Vegetation Richness")
```



#Logging has the highest amount of vegetation richness between land usage.

7. For which land use type (national parks, logging, or neither) is average all species diversity the greatest?

#park has the greatest average diversity of species

```
?mean
colnames(gabon)
```

```
## [1] "transect_id"      "distance"
## [3] "hunt_cat"         "num_households"
## [5] "land_use"          "veg_rich"
## [7] "veg_stems"         "veg_liana"
## [9] "veg_dbh"           "veg_canopy"
## [11] "veg_understory"   "ra_apes"
## [13] "ra_birds"          "ra_elephant"
## [15] "ra_monkeys"        "ra_rodent"
## [17] "ra_ungulate"       "rich_all_species"
## [19] "evenness_all_species" "diversity_all_species"
## [21] "rich_bird_species" "evenness_bird_species"
## [23] "diversity_bird_species" "rich_mammal_species"
## [25] "evenness_mammal_species" "diversity_mammal_species"
```

```
gabon%>%
  group_by(land_use)%>%
  summarise(mean_div=mean(diversity_all_species, na.rm=T))%>%
  arrange(desc(mean_div))
```

```
## # A tibble: 3 x 2
##   land_use mean_div
##   <chr>      <dbl>
## 1 Park      2.43
## 2 Neither   2.36
## 3 Logging   2.23
```

```
gabon%>%
  select(land_use, diversity_all_species)%>%
  filter(land_use=="Park")%>%
  mean(diversity_all_species, na.rm=T)
```

```
## Warning in mean.default(., diversity_all_species, na.rm = T): argument is not
## numeric or logical: returning NA
```

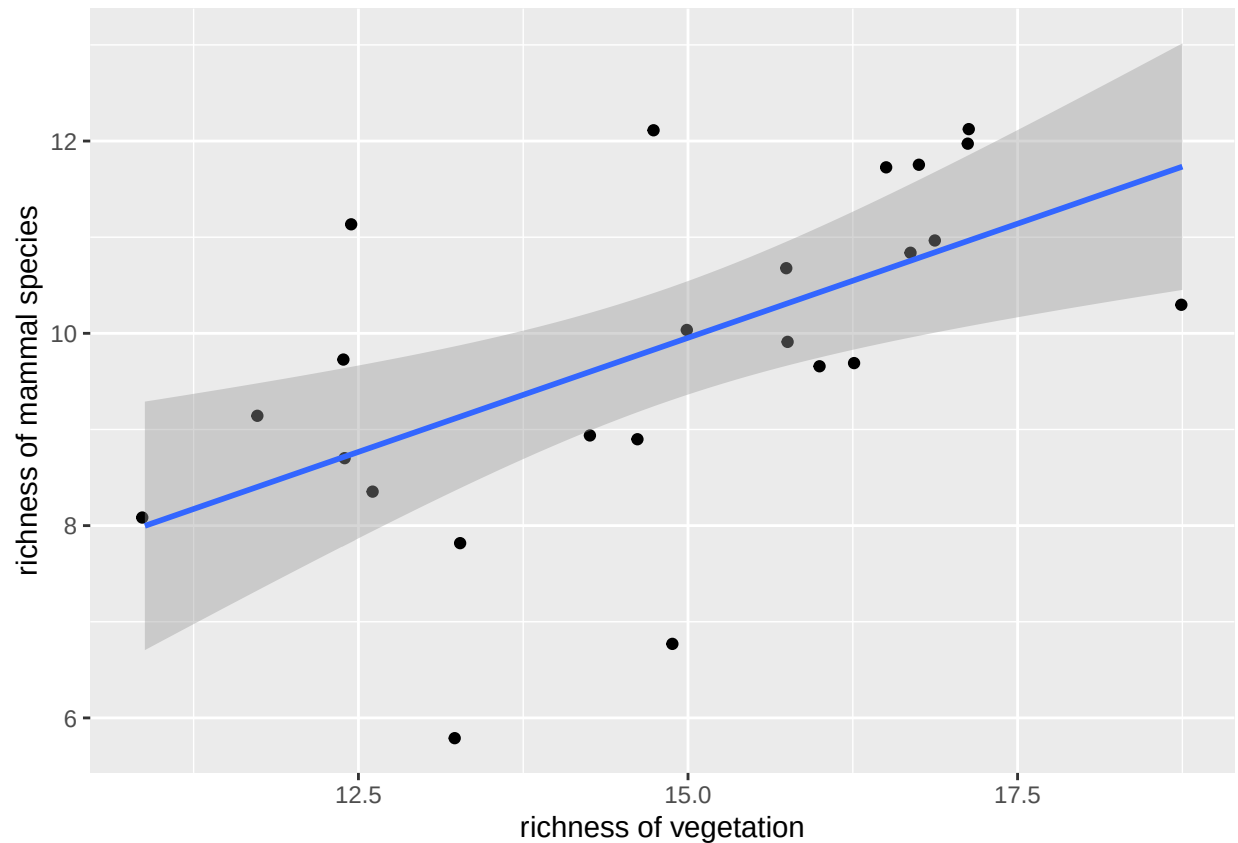
```
## [1] NA
```

```
#gabon%>%
#filter(diversity_all_species)
```

8. Is there a relationship between vegetation richness and mammal species richness?

```
ggplot(data=gabon,
  mapping=aes(x=veg_rich,y= rich_mammal_species))+
  geom_jitter()+
  geom_smooth(method = lm, se=T)+
  labs(x= "richness of vegetation", y= "richness of mammal species")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



#Yes, as the richness of vegetation increases, so does the richness of mammal species.

9. Use `filter` to find all transects that have greater than 15 tree species or a breast height diameter between 50 and 60cm.

```
colnames(gabon)
```

```
## [1] "transect_id"      "distance"
## [3] "hunt_cat"        "num_households"
## [5] "land_use"         "veg_rich"
## [7] "veg_stems"        "veg_liana"
## [9] "veg_dbh"          "veg_canopy"
## [11] "veg_understory"   "ra_apes"
## [13] "ra_birds"         "ra_elephant"
## [15] "ra_monkeys"       "ra_rodent"
## [17] "ra_ungulate"      "rich_all_species"
## [19] "evenness_all_species" "diversity_all_species"
## [21] "rich_bird_species" "evenness_bird_species"
## [23] "diversity_bird_species" "rich_mammal_species"
## [25] "evenness_mammal_species" "diversity_mammal_species"
```

```
filter(gabon, between(veg_dbh,50,60) | veg_rich>15)
```

```
## # A tibble: 14 x 26
```

```
##   transect_id distance hunt_cat num_households land_use veg_rich veg_stems
##   <dbl>      <dbl> <chr>          <dbl> <chr>      <dbl>    <dbl>
## 1         1        7.14 Moderate         54 Park        16.7     31.2
## 2         2       17.3   None           54 Park        15.8     37.4
## 3         2       18.3   None           29 Park        16.9     32.3
## 4         4       16.0   None           29 Park        17.1      36
## 5         5       17.5   None           29 Park        16.5     29.2
## 6         6       24.1   None           29 Park        14.8     31.2
## 7         9        5.13 High            73 Logging      16       27.1
## 8        14        8.23 Moderate        56 Logging      15.8     30.2
## 9        15        2.7   High            46 Neither      10.9     25.4
## 10       17        3.83 High            19 Neither      14.2     32.6
## 11       21        5.14 High            24 Neither      16.2     34.9
## 12       22        5.33 High            13 Logging      17.1     41.6
## 13       24       26.8   None           29 Park        16.8     37.3
## 14       26       11.2   Moderate        73 Logging      18.8     39.1
## # i 19 more variables: veg_liana <dbl>, veg_dbh <dbl>, veg_canopy <dbl>,
## #   veg_understory <dbl>, ra_apes <dbl>, ra_birds <dbl>, ra_elephant <dbl>,
## #   ra_monkeys <dbl>, ra_rodent <dbl>, ra_ungulate <dbl>,
## #   rich_all_species <dbl>, evenness_all_species <dbl>,
## #   diversity_all_species <dbl>, rich_bird_species <dbl>,
## #   evenness_bird_species <dbl>, diversity_bird_species <dbl>,
## #   rich_mammal_species <dbl>, evenness_mammal_species <dbl>, ...
```

10. Which transects and land use types have more than 10 tree species and 10 mammal species? Use `arrange()` to sort by the number of tree species.

```
filter(gabon, veg_rich>10 & (rich_mammal_species>10))%>%
arrange(desc(veg_rich))
```

```
## # A tibble: 9 x 26
##   transect_id distance hunt_cat num_households land_use veg_rich veg_stems
##   <dbl>      <dbl> <chr>          <dbl> <chr>      <dbl>    <dbl>
## 1         4       16.0   None           29 Park        17.1      36
## 2        22        5.33 High            13 Logging      17.1     41.6
## 3         2       18.3   None           29 Park        16.9     32.3
## 4        24       26.8   None           29 Park        16.8     37.3
## 5         1        7.14 Moderate        54 Park        16.7     31.2
## 6         5       17.5   None           29 Park        16.5     29.2
## 7        14        8.23 Moderate        56 Logging      15.8     30.2
## 8         6       24.1   None           29 Park        14.8     31.2
## 9         3       20.8   None           29 Logging      12.4     29.4
## # i 19 more variables: veg_liana <dbl>, veg_dbh <dbl>, veg_canopy <dbl>,
## #   veg_understory <dbl>, ra_apes <dbl>, ra_birds <dbl>, ra_elephant <dbl>,
## #   ra_monkeys <dbl>, ra_rodent <dbl>, ra_ungulate <dbl>,
## #   rich_all_species <dbl>, evenness_all_species <dbl>,
## #   diversity_all_species <dbl>, rich_bird_species <dbl>,
## #   evenness_bird_species <dbl>, diversity_bird_species <dbl>,
## #   rich_mammal_species <dbl>, evenness_mammal_species <dbl>, ...
```

Knit and Upload

Please knit your work as an .html file and upload to Canvas. Homework is due before the start of the next lab. No late work is accepted. Make sure to use the formatting conventions of RMarkdown to make your report neat and clean!